

HRUTHIK JAYARAMA

Portland, OR | hruthik@pdx.edu | +1 (503)-421-9673 | [LinkedIn](#)

Master's student in Electrical and Computer Engineering at Portland State University, specializing in Design Verification. Proficient in SystemVerilog, UVM, and SVA with hands-on experience in ASIC pre-silicon functional verification, constraint-random stimulus, coverage-driven verification, and UVM testbench development. Skilled in RTL bug identification through waveform analysis, scoreboard-driven debug, and driving verification sign-off at block-level and chip-level.

EDUCATION

Portland State University , Portland, OR <i>M.S., Electrical and Computer Engineering GPA: 3.75/4.0</i>	January 2025 - March 2027
Sai Vidya Institute of Technology (VTU University) , Bangalore, India <i>B.E., Electronics and Communication Engineering GPA: 3.4/4.0</i>	December 2020 - June 2024

SKILLS

Hardware Description / Verification Languages: SystemVerilog, Verilog, UVM, Python, C
Verification Methodologies: Functional Verification, Assertion-Based Verification (ABV), Constraint-Random Stimulus, Coverage-Driven Verification (CDV), Gate-Level Simulation, Pre-Silicon Verification, X-Propagation Analysis
UVM Components: UVM Agents, Drivers, Monitors, Sequencers, Scoreboards, Coverage Groups, Sequences, Virtual Sequences, Register Abstraction Layer (RAL)
EDA Tools: Siemens QuestaSim, Cadence Xcelium, ModelSim, MobaXterm, Xilinx Simulator
Verification Concepts: SVA Concurrent & Immediate Assertions, Functional & Code Coverage, Coverage Closure, FSM Verification, UVM Factory & Configuration, OOP Methodology
Protocols & Interfaces: UART, I2C, SPI, APB, AXI, MESI, MSI
Computer Architecture: Cache Coherence, Virtual Memory, Pipelining, Out-of-Order Execution, Branch Prediction, MIPS ISA
Operating Systems: Linux, Windows

PROJECTS

- Tomasulo's Algorithm – Cycle-Accurate Out-of-Order Execution Simulator** October 2025
- Implemented a cycle-accurate Tomasulo simulator with reservation stations, register renaming, and CDB to model dynamic out-of-order instruction scheduling and instruction latencies.
 - Validated hazard-free out-of-order execution with correct in-order commit against hand-traced references; demonstrated WAR/WAW hazard elimination through register renaming.
- APB Slave – Block-Level Functional Verification** July 2025
- Developed a modular UVM testbench for an APB slave DUT including a reusable APB agent (driver, monitor, sequencer), protocol-aware scoreboard, and functional coverage groups targeting all APB transfer types and error conditions.
 - Executed constraint-random regression in QuestaSim; achieved 100% functional and toggle coverage closure; gained hands-on proficiency in UVM RAL, bind-file assertions, and protocol-level debug methodology.
- Asynchronous FIFO – UVM Functional Verification** April 2025
- Designed a parameterized asynchronous FIFO with configurable depth and status flags (full, empty, almost-full); built a fully reusable UVM environment including driver, monitor, sequencer, scoreboard, and coverage groups with constraint-random stimulus generation.
 - Achieved full functional and code coverage closure in QuestaSim regression with zero data errors; demonstrated UVM component architecture, factory overrides, and coverage-driven verification methodology.
- MIPS-Lite 5-Stage Pipeline Simulator – Functional and Timing Simulation** April 2025
- Built a functional and timing simulator for a MIPS-lite ISA with a 5-stage pipeline (IF, ID, EX, MEM, WB), RAW hazard detection, branch misprediction handling, and configurable data forwarding.
 - Generated trace files to verify correctness and quantified IPC speedup with data forwarding versus stall-only hazard resolution; validated hazard classification and forwarding path behavior against hand-traced references.
- Serial Peripheral Interface (SPI) – Block-Level Functional Verification** January 2025
- Designed SPI master/slave controllers with FSM-driven shift registers; developed a comprehensive directed and constrained-random SystemVerilog testbench in QuestaSim covering all CPOL/CPHA modes and multi-slave full-duplex operation.
 - Resolved timing boundary violations via waveform analysis and scoreboard-driven debug, confirming correct data flow and clock-edge alignment across all configurations; gained expertise in protocol-level verification planning.
- Cache Controller – Split L1 / Shared L2 (MESI Coherence Protocol)** January 2025
- Designed a 4-core split L1 cache (4-way instruction, 8-way data, LRU) with a shared inclusive L2; implemented the full MESI coherence protocol for multi-core cache state management.
 - Verified all MESI state transitions under complex multi-core scenarios; tracked cache hit/miss performance metrics and gained deep understanding of coherence FSM verification and LRU replacement logic.

CERTIFICATIONS

Essential SystemVerilog for UVM – Cadence System Design
System Verilog for Design and verification – Cadence System design